Section 18 Solution

Bruno Smaniotto and Anson Tam*

3 Exercise

Initially there is one firm in a market for cars. The firm has a linear cost function: $C(q) = 2q$. The market inverse demand function is given by $P(Q) = 9 - Q$.

1. What price will the firm charge? What quantity of cars will the firm sell? How much profit will the firm make?

   Answer:

   The first two questions are about the case of monopoly. A monopolistic firm maximizes profits:

   $$\max_q \, (9 - q)\, q - 2q.$$

   and the first order condition is

   $$9 - 2q_M^* - 2 = 0.$$

   It follows that

   $$q_M^* = \frac{9-2}{2} = 7/2$$

   and

   $$p_M^* = 9 - q_M^* = 9 - 7/2 = 11/2.$$

   As for the profits in monopoly, they equal

   $$\pi_M^* = (9 - q_M^*)\, q_M^* - 2q_M^* = (9 - 7/2)\, 7/2 - 7 = 77/4 - 7 = 49/4$$

2. Now, a second firm enters the market. The second firm has an identical cost function. Suppose the firms compete via Bertrand competition. What will be the equilibrium price? What will be the equilibrium output of each firm? What sill be the equilibrium profit of each firm?

   $$\begin{aligned} p &= \frac{\partial C(q)}{\partial q} \\ p &= 2 \\ \Rightarrow P(Q) &= 9 - Q \\ 2 &= 9 - Q \\ Q &= 7 \\ \Rightarrow q &= 7/2 \\ \Rightarrow \pi &= 0 \end{aligned}$$

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3. Suppose instead the firms compete via Cournot competition. What will be the equilibrium output of each firm? What will be the equilibrium price? What will be the equilibrium profit of each firm?

   Answer:

   We are now in a Cournot competition set-up. A duopolistic firm maximizes profits:

$$\max_{q_i} \left(9 - (q_i + q_{-i})\right) q_i - 2q_i.$$

   It follows that the first order conditions for the two firms are

$$9 - 2q_1^* - q_2^* - 2 = 0 \qquad\qquad 9 - 2q_2^* - q_1^* - 2 = 0$$

   Solving this system of two equations we will get

$$q_1^* = q_2^* = \frac{7}{3}$$

   Not surprisingly, the quantities produced by firms 1 and 2 are equal.

   The total market output $Q^* = 2 * 7/3 = 14/3$ and therefore the market price is $p(Q) = 9 - 14/3$.

   In the Cournot case, the profit for either firm is

$$\pi_C = (9 - 14/3)\, 7/3 - 14/3 = 91/9 - 14/3 = 49/9.$$

4. Which type of market do consumers prefer: Monopoly, Bertrand duopoly, or Cournot duopoly? Why?

   Answer:

   Consumers prefer the market with the most total output (and by consequence the lowest price), since this maximizes consumer surplus. Formally, consumers maximize

$$\int_0^Q \left(D\left(p\right) - p\right) dq' = \int_0^Q \left((9 - q') - (9 - Q)\right) dq' = \left[Qq' - q'^2/2\right]_0^Q = Q^2 - Q^2/2 = Q^2/2,$$

   which is increasing in total quantity produced. (This integration was not required.) Therefore consumers prefer the Bertrand duopoly, which has the highest total production.